
MODEL SELECTION AND VALIDATION REPORT (haul_duration model)

OBSERVATIONS:3451

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (haul_duration model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: gaussian(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	11.11	2.31
haul_dur	0.05	0.03

Dispersion parameter: 2.42

Spatiotemporal AR1 correlation (rho): 0.98

Matérn range: 553.37

Spatiotemporal marginal AR1 SD: 4.34

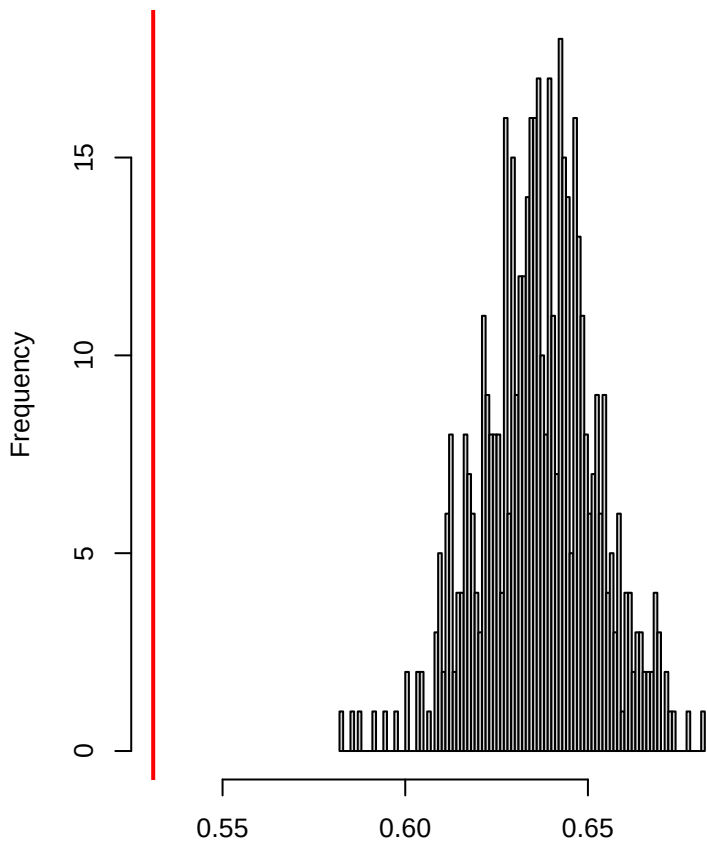
REML criterion at convergence: 8270.701

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

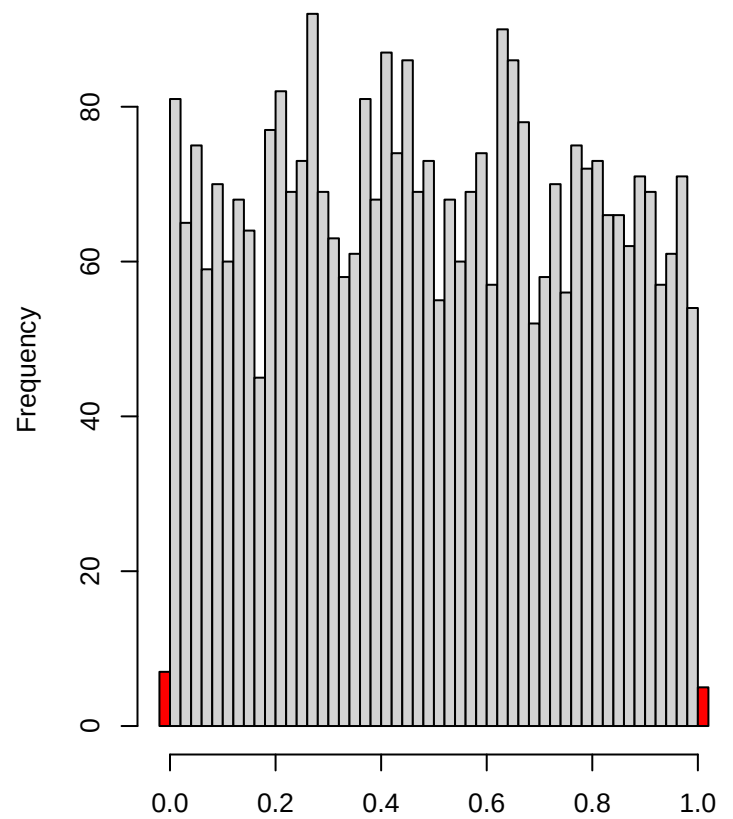
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	6	16553.40	3.077348
spatiotemporal_random_fields_ar1_with_anisotropy	8	16556.48	462.656667
spatiotemporal_random_fields_iid	5	17019.14	1.257883
spatiotemporal_random_fields_iid_with_anisotropy	7	17020.39	62.530343
spatial_random_fields	5	17082.92	3.765407
spatial_random_fields_with_anisotropy	7	17086.69	640.116743
error_only	3	17726.81	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



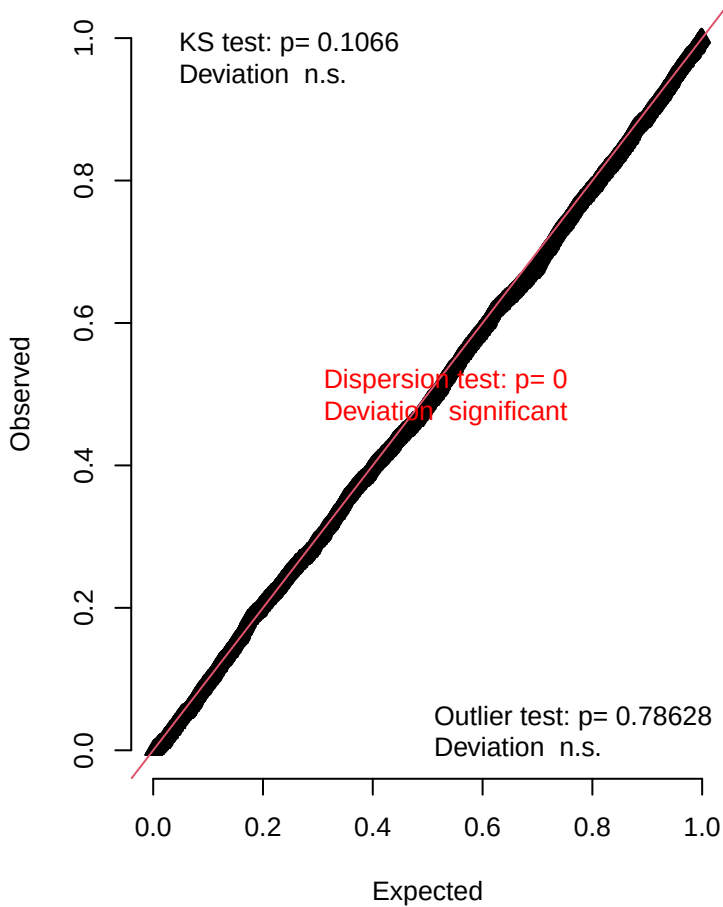
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test n.s.

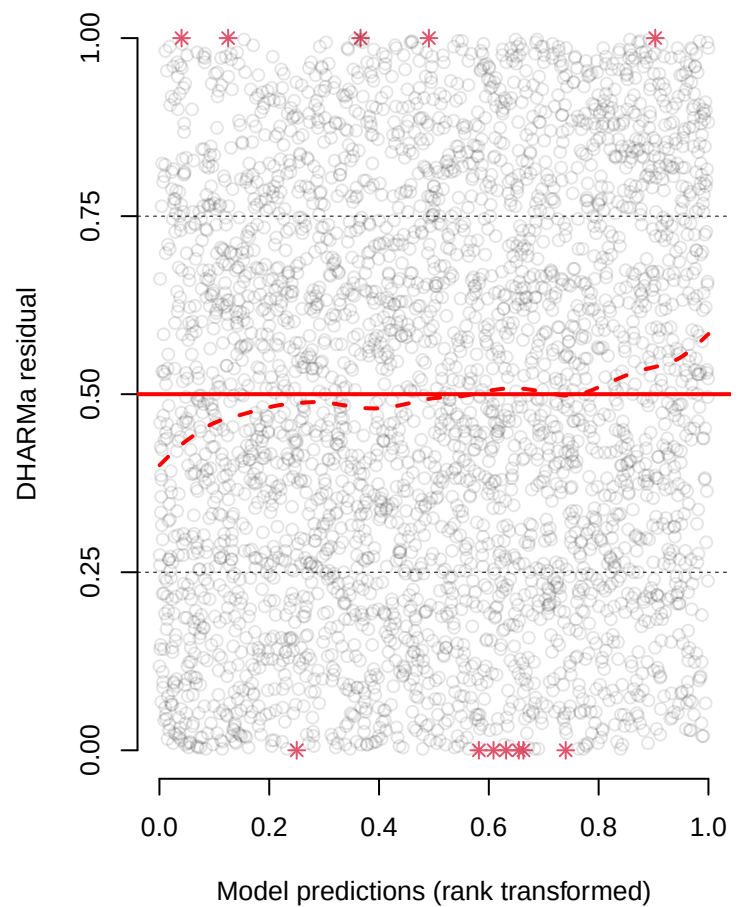


Residuals (outliers are marked red)

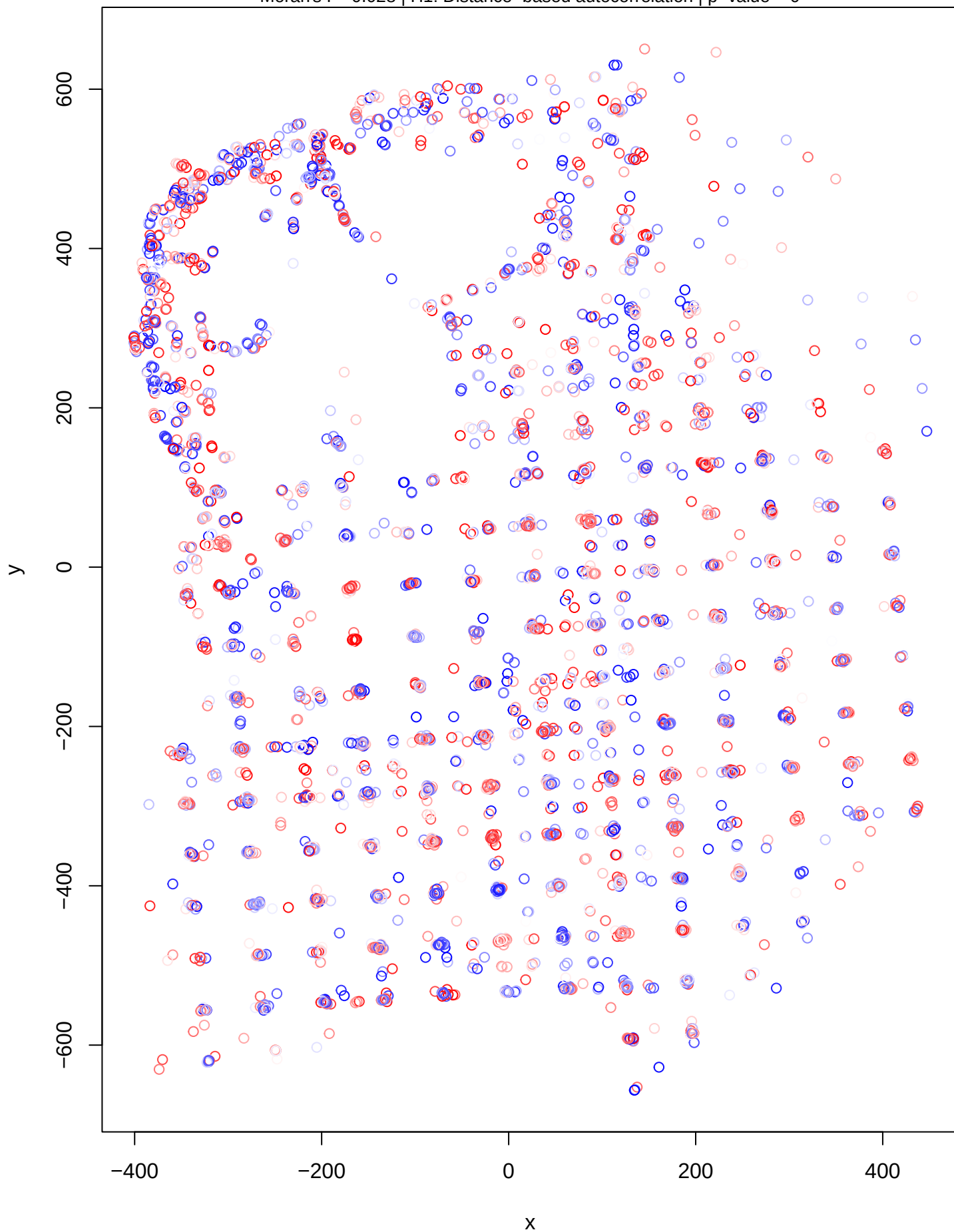
QQ plot residuals



DHARMa residual vs. predicted



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.028 | H1: Distance-based autocorrelation | p-value = 0



DHARMa residual Residual vs. pred

Predictor: haul_dur

